BrandGuard AI – Intelligent Governance Platform

Inkloom AI Hackathon Round 2 Build Report • The Build Phase • October 2026

Project: BrandGuard AI	**Target Stack:** React 18, Flask, Gemini 2.5
Repository: github.com/your-username/brandguard-ai	**Deployment:** Vercel (Front) + Render (API)
Verification: 19/19 Unit/E2E Tests Passed	**Status:** Production Ready (Phases 1–5 Complete)

1. Problem Statement

Enterprise marketing teams and creative agencies face acute brand safety risks when drafting high-velocity promotional content. The proliferation of generative AI tools has accelerated content output, but has dramatically elevated the risk of unsubstantiated claims (e.g., '100% downtime-proof guarantee', 'zero cost forever'), off-brand tone, and hallucinated organizational facts. Manual review cycles in spreadsheets lack audit history, versioning snapshots, and objective rule enforcement.

2. Proposed Solution

BrandGuard AI is a multi-tenant brand governance platform delivering an end-to-end, human-in-the-loop compliance engine:

- **Centralized Brand Rulebooks:** Full governance across 6 categories (Tone, Terminology, Messaging, Approved Facts, Restricted Claims, Accessibility).
- **Evidence-Grounded AI Auditing:** Google Gemini 2.5 Flash evaluates copy against active guidelines, providing line quotes, severity ratings, and replacement text.
- **Immutable Revision Snapshots:** Every content update creates an unalterable revision with changelog reasons and word count tracking.
- **Stale Report Protection:** Content revisions automatically invalidate prior AI analyses, preventing outdated reports from being submitted for approval.
- **Manager Sign-Off & Audit Trail:** Content locks during review. Managers sign off or request revisions with required directives; all actions are logged in ReviewDecision.

3. Completed & Verified Features (Phases 1–5)

Feature Area	Key Capabilities & Verification
Authentication & RBAC	Stateless JWT auth with roles for Content Creator and Brand Manager. Secure password hashing.
Brand Rulebooks	6-category taxonomy, team membership isolation, approved facts, and prohibited keyword registries.
Content Drafting & History	Rich draft editor, word/char counters, immutable ContentRevision snapshots, and version preview modal.
Gemini AI Engine	Official google-genai SDK, structured JSON output, rule ID sanitization, severity counters.
Manager Approval Workflow	Approval queue (/manager/approvals), warning acknowledgement modal, status locking, ReviewDecision timeline.
Production Deployment	Gunicorn WSGI, PostgreSQL connection pooling, Vercel SPA rewrites, safe logging, non-destructive seeder.

4. Automated Testing & Verification Evidence

BrandGuard AI enforces zero-regression test coverage across all authorization boundaries and workflow transitions:

Test Suite File	Tests	Scope & Validated Invariants	Status
test_auth.py	6	Registration, password hashing, token expiration, RBAC	100% PASS
test_phase2.py	3	Brand CRUD, guideline taxonomy, revision snapshotting	100% PASS
test_phase3.py	5	Gemini integration, rule sanitization, stale-report detection	100% PASS
test_phase4.py	2	Manager queue, warning acknowledgement, decision log	100% PASS
test_production_workflows.py	3	Cross-user isolation, status locking, full E2E lifecycle	100% PASS

5. Planned Round 3 Roadmap Innovations

1. **Multimodal Visual & Ad Banner Compliance:** Ingest PNG/JPG display ads and PDFs with Gemini multimodal vision to audit logo placement, contrast ratios, and legal disclaimer legibility.
2. **Advanced Governance Analytics:** Cross-campaign dashboards tracking violation frequency, creator velocity, and team compliance trends over time.
3. **Autonomous Web Verification & Certificates:** Web-grounded citation checking of customer and benchmark metrics, exporting cryptographically signed PDF audit certificates.

6. Evaluator / Judge Quickstart

A non-destructive seeder is included for instant review (run python seed_demo.py in BACKEND):
• **Brand Manager:** judge.manager@brandguard.ai / DemoPassword123!
• **Content Creator:** judge.creator@brandguard.ai / DemoPassword123!
• Pre-loaded Brand: **CloudNova** (with 6 guidelines, violation draft, and pending approval ad).

BrandGuard AI • Inkloom AI Hackathon Round 2 Build • Verified Complete & Deployment Ready